

Jharkhand University of Technology, Ranchi**Diploma 2nd Semester Examination, 2024 (NEP)****Subject : Statistics And Analytics****Subject Code : AEC201****Time Allowed : 3 Hours****Full Marks : 70***Answer in your own words.**Answer any five questions.**Question No. 1 is compulsory, and from rest of
the questions answer any four only.**All questions carry equal marks.***1. Objective type questions, pick the right options:****2×7=14**

- (i) Which graph is best for showing how something is divided into parts?
- (a) Line graph (b) Pie chart
(c) Histogram (d) Scatter plot
- (ii) What is the mode of the following dataset: 3, 5, 7, 7, 9?
- (a) 3 (b) 5
(c) 7 (d) 9
- (iii) How can you display values directly on a chart in Excel?
- (a) Add a Trendline (b) Use Data Labels
(c) Insert a Table (d) Apply Conditional Formatting
- (iv) What is the purpose of a legend in an Excel chart?
- (a) To display the chart title (b) To explain the data series
(c) To show the axis labels (d) To calculate values
- (v) What is the output of the following code?
- Print (2 * 3 ** 2)
- (a) 36 (b) 18
(c) 12 (d) 9
- (vi) Which of the following data types is immutable in Python?
- (a) List (b) Dictionary
(c) Set (d) Tuple
- (vii) Which function is used to read input from the user in Python?
- (a) read() (b) input()
(c) scan() (d) enter()

2. (a) Define and give an example of qualitative data, quantitative data, discrete data, and continuous data. How do they differ?
- (b) Explain what a histogram is and describe how it is used to represent data. Describe the different shapes a histogram can take and what they indicate about the data distribution. 7+7
3. (a) The heights (in cm) of 7 plants are recorded as: 25, 30, 28, 25, 32, 30, 28. Determine the range, mean, median, and mode of the plant heights using Excel.
- (b) The dataset of types of movies watched in a month is: Action (40 movies), Comedy (30 movies), Drama (20 movies), Horror (10 movies). Create a pie chart in Excel to visualize this data and calculate the percentage of each movie type. 7+7
4. (a) The number of apples sold each day over a two-week period is recorded as: 10, 12, 10, 14, 12, 10, 12, 15, 14, 10, 12, 15, 10, 12, 14. Construct a frequency table to display the number of apples sold.
- (b) For the dataset of average temperatures in three cities (City A, City B, City C) over four months: January (30, 25, 28), February (32, 26, 29), March (35, 28, 30), April (40, 30, 32), create a grouped bar graph. Determine the highest average temperature for each city. 7+7
5. (a) The following data represents the ages of a group of people: 18, 22, 24, 26, 30, 34, 38. Determine the skewness and kurtosis of this dataset.
- (b) Define Array. What is the purpose of the array module in Python? How do you create an array of float numbers using the array module in Python? 7+7
6. (a) How do you define a function in Python? Write a program in python to print the Cube of a number using function.
- (b) Define operator in Python. Discuss different types of operators in detail with example. 7+7
7. Write short notes on *any four* of the following: 3.5×4
- (a) If-else statement
 - (b) While loop
 - (c) Standard deviation
 - (d) Variance
 - (e) Line Graph
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